Segment 1A

Introduction to Microcomputer and Microprocessor

1.1 General Architecture of a Microcomputer System:

The term *microcomputer* is generally synonymous with personal computer, or a computer that depends on a microprocessor. Microcomputer is used to describe a system that includes a minimum of a microprocessor, program memory, data memory, and input/output (I/O). Some microcomputer systems include additional components such as timers, counters, analogue-to-digital converters and so on. Thus, a microcomputer system can be anything from a large computer having hard disks, floppy disks and printers, to a single chip computer system.
The hardware of a microcomputer system can be divided into four functional sections:

(a) Input unit
(b) Microprocessor Unit
(c) Memory Unit, and
(d) Output Unit.

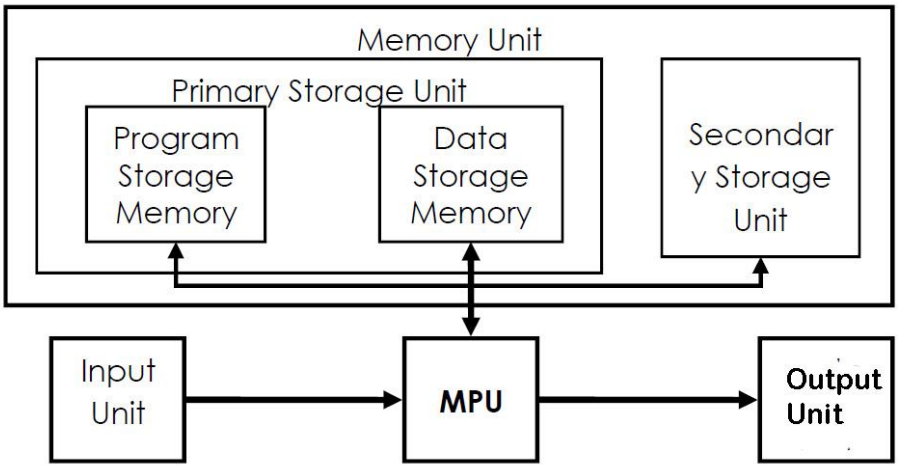


Figure 1: General Architecture of a Microcomputer System

Input Unit and Output Unit: Input and Output units are the means by which the MPU communicates with the outside world.

Input unit: keyboard, mouse, scanner etc.

Output unit: Monitor, printer etc.

Bi-directional ports: Modem, Network etc.

I/O is responsible for interfacing between CPU and operators (or external devices). I/O includes any device that converts binary data in the computer into another form of data that are used by human beings. Examples of I/O devices include monitors, keyboards, printers, modems, mice, hard disks, floppy disks, etc. Each computer system may have many I/O ports. Each input port

has a unique address and each output port has a unique address. Some microprocessors assign the memory and I/O with separate address space (e.g., Intel x86 processors). In this case, one may use memory address X and also I/O address X the same time. Some microprocessors make the memory and I/O use the same address space (e.g., Motorola 68HC12). In this case, if memory uses address X, I/O cannot use address X.

Microprocessor Unit/ Central Processing Unit:

CPU has two main components

(a) Arithmetic and logic unit (ALU) :Responsible for computation only

(b) Control unit:

- Responsible for sending and gathering information from memory and I/O
- Responsible for coordinating the sequence and timing of the activities on the whole system.

CPU also has several storage places called registers.

CPU is the "brain" of the microcomputer. It provides the decision making ability for the computer. CPU has two parts, the arithmetic and logic unit (ALU), and control unit. ALU is responsible for arithmetic and logic operations. Control unit is responsible for coordinating the data flow and to control the sequence and timing of circuit operations in the system. Registers are a very small but fast accessing storage space that holds very essential information for program execution and decision making. In program execution, the CPU reads and executes the programs instructions one by one from the main memory. The execution of instructions may involve the arithmetic/logic operations and/or transfer data between CPU and main memory (or I/O ports).

Memory Unit:

(a) Processor Memory

(b) Primary (or main) memory

(c) Secondary memory

- Hard disks, CD, floppy disks, tape

Memory is responsible for storing programs and the data that are needed by the program. To make the concept clear, we divide the computer memory to processor memory, main memory (or primary memory) and secondary memory.

Processor Memory refers to a set of CPU registers. These registers are useful to hold temporary results when a computation is in progress. Also, there is no speed disparity between the registers and the microprocessor because they are fabricated using the same technology. The main disadvantage is the cost involved which forces the architect to include very few registers (usually 8 to 16 only) in the microprocessor.

The *main memory* is the memory that the CPU can access directly. Examples of main memory include RAM, ROM, etc.

The *secondary memory* cannot be addressed directly (cannot access specified memory location) by the CPU. Examples of secondary include floppy disk, hard disk, CD, etc. The information in the secondary memory must be copied to the main memory so that CPU can access it. Secondary memory is much cheaper than primary memory.

1.2 Microprocessor and Microcontroller

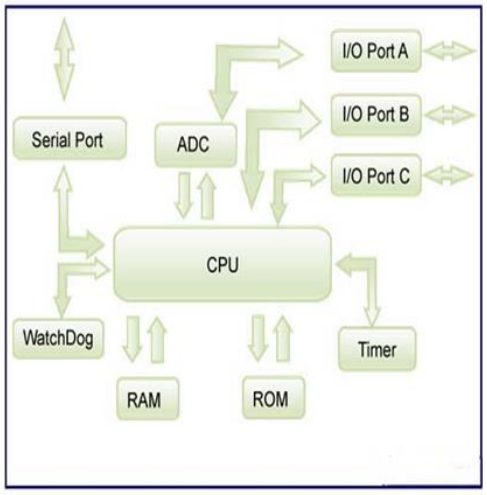
Microprocessor is a multipurpose, programmable register based electronic device which read binary instructions from memory, processes the input data as per instructions and provides output. It is an IC which has only the CPU inside them i.e. only the processing powers such as Intel's Pentium 1,2,3,4, core 2 duo, i3, i5 etc. These microprocessors don't have RAM, ROM, and other peripheral on the chip. A system designer has to add them externally to make them functional.

Microcontroller is a device that includes microprocessor, memory and input/output devices on a single chip. Microcontroller has a CPU, in addition with a fixed amount of RAM, ROM and other peripherals all embedded on a single chip. At times it is also termed as a mini computer or a computer on a single chip.

1.3 Comparison of Microcontroller and Microprocessor

Microcontroller :

1. Microcontroller having inbuilt RAM or ROM and inbuilt timer.
2. Input and output ports are available.
3. Inbuilt serial port.
4. Separate memory to store program and data.
5. Many functional pins on the IC.
6. Boolean operation directly possible.
7. It takes few instructions to read and write data from external memory.



Microprocessor :

1. Do not have inbuilt RAM or ROM and timer.
2. Input and output ports are not available, requires extra device like 8255
3. Do not have inbuilt serial port, requires 8250 device.
4. Program and data are stored in same memory.
5. Less multifunction pins on IC.
6. Boolean operation is not possible directly.
7. It takes many instructions to read and write data from external memory.

1.4 Microprocessor Characteristics

Three basic characteristics that differentiate microprocessors are-

- **Instruction set**: The set of instructions that the microprocessor can execute.
- **Bandwidth**: The number of bits processed in a single instruction.
- **Clock speed**: Given in megahertz (MHz), the clock speed determines how many instructions per second the processor can execute.

In all cases, the higher the value, the more powerful the CPU. For example, a 32 bit microprocessor that runs at 50MHz is more powerful than a 16-bit microprocessor that runs at 25MHz.

1.5 Types of Microprocessors:

Microprocessors generally is categorized in terms of the maximum number of binary bits in the data they process (i.e. number of bits that their ALU can work with at a time) – that is, their word length. Over time, five standard data widths have evolved for microprocessors: 4-bit, 8-bit, 16-bit, 32-bit, 64-bit.
So, 16-bit microprocessor means that ALU can work with 16-bit number at a time or, data width of this microprocessor is 16-bit.
There are so many manufacturers of Microprocessors, but only two companies have been produces popular microprocessors: *Intel* and *Motorola*.

1.6 Evaluation of Microprocessor:

We can divide the years of development of microprocessors as 5 generations (based on data width):

First generation (1971 – 73)

Intel Corporation introduced 4004, the first microprocessor in 1971. It is evolved from the development effort while designing a calculator chip.
There were three other microprocessors in the market during the same period:

- Rockwell International's PPS-4 (4 bits)
- Intel's 8008 (8 bits)
- National Semiconductor's IMP-16 (16 bits)

They were fabricated using PMOS technology which provided low cost, slow speed and low output currents. It contained 2300 PMOS transistor.
They were not compatible with TTL.

Second Generation (1974 – 1978)

Marked the beginning of very efficient 8 – bit microprocessors.
Some of the popular processors were:

- ✓ Motorola's 6800 and 6809
- ✓ Intel's 8085
- ✓ Zilog's Z80

They were manufactured using NMOS technology.
This technology offered faster speed and higher density than PMOS
It is TTL compatible.

Third generation microprocessors (1979 – 80)

This age is dominated by 16 – bits microprocessors
Some of them were:

- ✓ Intel's 8086/80186/80286
- ✓ Motorolla's 68000/68010

They were designed using HMOS technology
HMOS provides some advantages over NMOS as Speed-power-product of HMOS is four times better than that of NMOS.
HMOS can accommodate twice the circuit density compared to NMOS

Intel used HMOS technology to recreate 8085A and named it as 8085AH with a higher price tag.

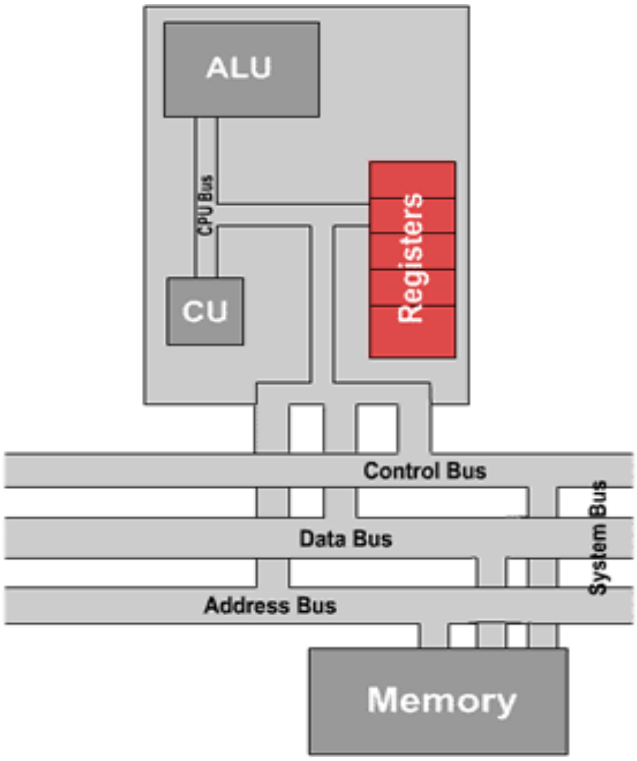
Fourth Generation (1981 – 1995)
This era marked the beginning of 32 bits microprocessors

- ✓ Intel introduced 432, which was bit problematic
- ✓ Then a clean 80386 in launched.
- ✓ Motorola introduced 68020/68030.

They were fabricated using low-power version of the HMOS technology called HCMOS.
Motorola introduced 32-bit RISC processors called MC88100

Fifth Generation (1995 – till date)
This age the emphasis is on introducing chips that carry on-chip functionalities and improvements in the speed of memory and I/O devices along with introduction of 64-bit microprocessors.
Intel leads the show here with Pentium, Celeron and very recently dual and quad core processors working with up to 3.5GHz speed.

1.7 Microprocessor Architecture



As there are a great many variations in architecture between the different kinds of CPU/MPU, we shall begin my looking at a simplified model of the structure. The simplified model consists of five parts, which are:

1.7.1 ***Arithmetic & Logic Unit (ALU)***

The part of the central processing unit that deals with operations such as addition, subtraction and multiplication of integers and Boolean operations. It receives control signals from the control unit telling it to carry out these operations. It works in conjunction with the register array for many of these, in particular, the accumulator and flag registers. The accumulator holds the results of operations, while the flag register contains a number of individual bits that are used to store information about the last operation carried out by the ALU.

Some of the tasks performed by the ALU are given below:

Addition and subtraction: These two tasks are performed by constructs of logic gates, such as half adders and full adders. While they may be termed 'adders', with the aid of they can also perform subtraction via use of inverters and 'two's complement' arithmetic.

Multiplication and division: In most modern processors, the multiplication and division of integer values is handled by specific floating-point hardware within the CPU. Earlier processors used either additional chips known as maths co-processors, or used a completely different method to perform the task.

Logical tests: Further logic gates are used within the ALU to perform a number of different logical tests, including seeing if an operation produces a result of zero. Most of these logical tests are used to then change the values stored in the flag register, so that they may be checked later by separate operations or instructions. Others produce a result which is then stored, and used later in further processing.

Comparison: Comparison operations compare values in order to determine such things as whether one number is greater than, less than or equal to another. These operations can be performed by subtraction of one of the numbers from the other, and as such can be handled by the aforementioned logic gates. However, it is not strictly necessary for the result of the calculation to be stored in this instance; the amount by which the values differ is not required. Instead, the appropriate status flags in the flag register are set and checked to determine the result of the operation.

Bit shifting: Shifting operations move bits left or right within a word, with different operations filling the gaps created in different ways. This is accomplished via the use of a shift register, which uses pulses from the clock within the control unit to trigger a chain reaction of movement across the bits that make up the word. Again, this is a quite complicated logical procedure, and further reading may aid your understanding.

1.7.2 ***Control Unit (CU)***

This controls the movement of instructions in and out of the processor, and also controls the operation of the ALU. It consists of a decoder, control logic circuits, and a clock to ensure everything happens at the correct time. It is also responsible for performing the instruction execution cycle. The three main elements of the control unit are as follows:

Decoder: This is used to decode the instructions that make up a program when they are being processed, and to determine in what actions must be taken in order to process them. These decisions are normally taken by looking at the opcode of the instruction, together with the addressing mode used.

Timer or clock: The timer or clock ensures that all processes and instructions are carried out and completed at the right time. Pulses are sent to the other areas of the CPU at regular intervals (related to the processor clock speed), and actions only occur when a pulse is detected. This ensures that the actions themselves also occur at these same regular intervals, meaning that the operations of the CPU are synchronized.

Control logic circuits: The control logic circuits are used to create the control signals themselves, which are then sent around the processor. These signals inform the arithmetic and logic unit and the register array what they actions and steps they should be performing, what data they should be using to perform said actions, and what should be done with the results.

1.7.3 ***Register Array****:*

This is a small amount of internal memory that is used for the quick storage and retrieval of data and instructions. All processors include some common registers used for specific functions, namely the program counter, instruction register, accumulator, memory address register and stack pointer. Many different types of registers are common between most microprocessor designs. These are:

Program Counter (PC): This register is used to hold the memory address of the next instruction that has to execute in a program. This is to ensure the CPU knows at all times where it has reached, that is able to resume following an execution at the correct point, and that the program is executed correctly.

Instruction Register (IR): This is used to hold the current instruction in the processor while it is being decoded and executed, in order for the speed of the whole execution process to be reduced. This is because the time needed to access the instruction register is much less than continual checking of the memory location itself.

Accumulator (A, or ACC): The accumulator is used to hold the result of operations performed by the arithmetic and logic unit

Memory Address Register (MAR): Used for storage of memory addresses, usually the addresses involved in the instructions held in the instruction register. The control unit then checks this register when needs to know which memory address to check or obtain data from.

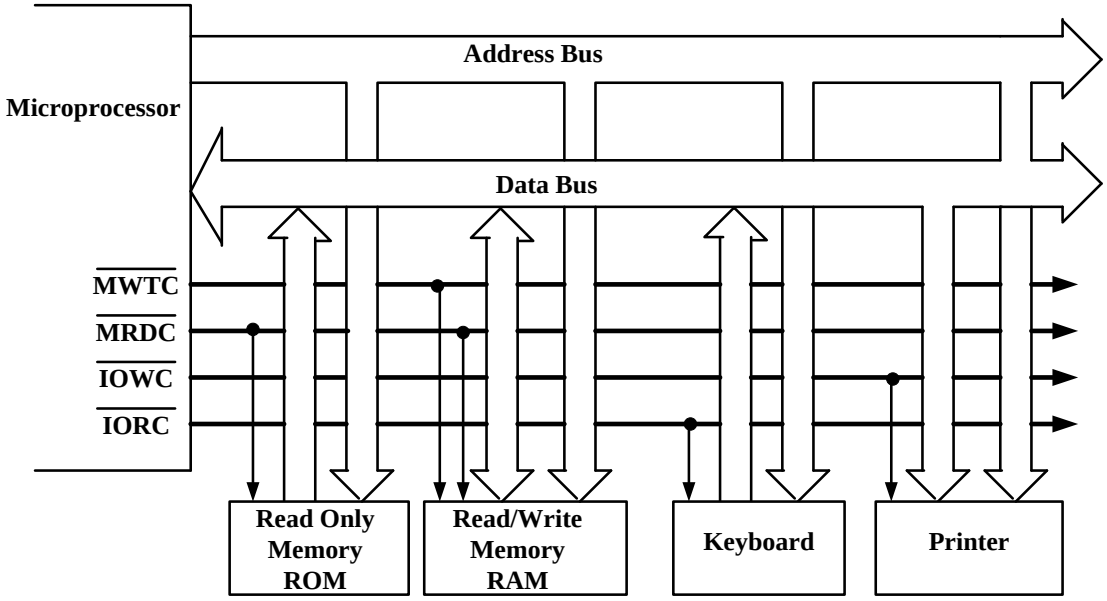
Memory Buffer Register (MBR): When an instruction or data is obtained from the memory or elsewhere, it is first placed in the memory buffer register. The next action to take is then determined and carried out, and the data is moved on to the desired location.

Flag register / status flags: The flag register is specially designed to contain all the appropriate 1-bit status flags, which are changed as a result of operations involving the arithmetic and logic unit.

Other general purpose registers: These registers have no specific purpose, but are generally used for the quick storage of pieces of data that are required later in the program execution. In the model used here these are assigned the names A and B, with suffixes of L and U indicating the lower and upper sections of the register respectively.

1.7.4 **System Bus:**

It is used for connections between the processor, memory and peripherals, and transferal of data between the various parts. The system bus consists of three different groups of wiring, called the data bus, control bus and address bus. These all have separate responsibilities and characteristics, which can be outlined as follows:



Control Bus: The control bus contains lines that select the memory or I/O and cause them to perform a read or write operation. In most computer system, there are four control bus connections:

- $\overline{MRDC}$ (Memory read control)
- $\overline{MWTC}$ (Memory Write control)
- $\overline{IORC}$ (I/O read control)
- $\overline{IOWC}$ (I/O write control)

Data Bus: This is used for the exchange of data between the processor, memory and peripherals, and is bi-directional so that it allows data flow in both directions along the wires. Data bus may be 8 bits, 16 bits, 32 bits or 64 bits.

Address Bus: This is a unidirectional bus. This bus is usually 8 to 32 bits wide. Information transfer takes place from the microprocessor to the memory or I/O elements. For a 16 bit address bus, microprocessor can generate 2^{16} = 65,536 different possible address. Each one of these addresses represents a definite memory location or an I/O element.

1.7.5 Memory:

The memory is not an actual part of the CPU itself, and is instead housed elsewhere on the motherboard. However, it is here that the program being executed is stored, and as such is a crucial part of the overall structure involved in program execution.

1.8 Instruction Execution

To execute an instruction processor must follows minimum of four steps:

(1) Instruction Fetch
(2) Instruction Decode
(3) Instruction execute
(4) Store or, Store of Result

Instruction Fetch*:* Control unit collect the instructions from main memory and put them in CPU register. This is called instruction fetch.

Instruction Decode: When instruction reaches in processor register, CU decodes or interprets the instruction and sends necessary signals and data to ALU.

Instruction Execute: ALU process the data with arithmetic and logic operations and gives a result according to instructions.

Store Result: Finally CU stores result in Accumulator or main memory.

These four steps to execute an instruction are called machine cycle.

1.9 Microprocessor Architecture (Based on Instruction Set):

As microprocessor is a programmable device, obviously it has an instruction set to control its operation by programming. There are two prevalent instruction set architectures:

- ✓ RISC (Reduced Instruction Set Computer) Architecture.
- ✓ CISC (Complex Instruction Set Computer) Architecture.

RISC	CISC
Small number of instructions in the instruction set compared to CISC.	CISC chips have a large amount of different and complex instructions
RISC chips use simpler instructions sets to achieve higher clock frequencies and process more instructions per clock cycle than CISC processors	In common, CISC chips process smaller instructions per clock cycle than RISC processors, but to do same task number of instruction is smaller in CISC than RISC.
It's hardware design is easy but software (programming) is difficult because you have to write code using a small number of instructions.	It's hardware design is difficult but programming is easy as it has a large number of instructions.
Apple and SUN use RISC architecture.	Intel and AMD develop CISC processors. 8086 microprocessor has CISC architecture.

Typically, CISC chips have a large amount of different and complex instructions. The philosophy behind it is that hardware is always faster than software, therefore one should make a powerful instruction set, which provides programmers with assembly instructions to do a lot with short programs.

There is still considerable controversy among experts about which architecture is better. Some say that RISC is cheaper and faster and therefor the architecture of the future. Others note that by making the hardware simpler, RISC puts a greater burden on the software. Software needs to become more complex. Software developers need to write more lines for the same tasks. Therefore they argue that RISC is not the architecture of the future, since conventional CISC chips are becoming faster and cheaper anyway.